

INTEGRALES IRRACIONALES.

$$241) \int \frac{\sqrt{x}}{\sqrt[4]{x^3+1}} dx =$$

Transformamos las raíces en exponentes:

$$\int \frac{x^{\frac{1}{2}}}{x^{\frac{3}{4}} + 1} dx = (*)$$

Obtenemos min. común múlt. de los denominadores de los exponentes:

$$\frac{1}{2} \text{ y } \frac{3}{4} \rightarrow \text{m.c.m. de los denominadores: 4}$$

Sustituimos "x" por "t" y elevamos a "t" al exponente que resulte del m.c.m.

$$\text{En este caso: } x = t^4$$

$$dx = 4t^3 dt$$

$$(*) = \int \frac{(t^{\frac{1}{2} \cdot 4})}{(t^{\frac{3}{4} \cdot 4} + 1)} 4t^3 dt = 4 \int \frac{t^2}{t^3 + 1} \cdot t^3 dt = 4 \int \frac{t^5}{t^3 + 1} dt =$$

PUEDEN RESOLVER EL COCIENTE Y UTILIZAR CUALQUIERA DE LOS MÉTODOS DE INTEGRACIÓN

$$242) \int \frac{\sqrt{x+4}}{x} dx =$$

$$243) \int \frac{\sqrt{x^3} - \sqrt[3]{x}}{6\sqrt[4]{x}} dx =$$

RESOLUCIÓN DE EJERCICIOS DE INTEGRALES IRRACIONALES.

$$241) \quad 4 \int \frac{t^5}{t^3+1} dt = 4 \int \left(t^2 - \frac{t^2}{t^3+1} \right) dt = 4 \left[\int t^2 dt - \int \frac{t^2}{t^3+1} dt \right]$$

$$\boxed{\frac{t^5}{t^3+1} = t^2 - \frac{t^2}{t^3+1}}$$

$$= 4 \left[\frac{1}{3} t^3 - \int \frac{t^2}{t^3+1} dt \right] = \textcircled{1}$$

* RESOLVEMOS
* POR SUSTITUCIÓN

$$* \int \frac{t^2}{t^3+1} dt = \int \frac{t^2}{u} \left(\frac{du}{3t^2} \right) = \frac{1}{3} \int \frac{du}{u} =$$

$$\begin{aligned} u &= t^3+1 \\ du &= 3t^2 dt \\ dt &= \frac{du}{3t^2} \end{aligned}$$

$$= \frac{1}{3} \ln u + C = \frac{1}{3} \ln(t^3+1) + C.$$

$$\textcircled{1} = 4 \left[\frac{1}{3} t^3 - \frac{1}{3} \ln(t^3+1) + C \right] = \frac{4}{3} t^3 - \frac{4}{3} \ln(t^3+1) + C.$$

Si consideramos que $x = t^4 \rightarrow t = x^{\frac{1}{4}} \rightarrow t = \sqrt[4]{x}$.

$$= \frac{4}{3} (\sqrt[4]{x})^3 - \frac{4}{3} \ln(\sqrt[4]{x})^3 + 1 + C.$$

$$242) \int \frac{\sqrt{x+4}}{x} dx = \int \frac{(x+4)^{\frac{1}{2}}}{x} dx$$

exponente: $\frac{1}{2}$

m.c.m.: 2

$$\boxed{x+4 = t^2}$$

$$\text{si } x+4 = t^2$$

$$x = t^2 - 4$$

$$dx = 2t dt$$

$$= \int \frac{(t^2)^{\frac{1}{2}}}{t^2-4} \cdot 2t dt =$$

$$= \int \frac{t}{t^2-4} \cdot 2t dt = 2 \int \frac{t^2}{t^2-4} dt = *$$

$$\frac{t^2}{t^2-4} = 1 + \frac{4}{t^2-4}$$

* POR DESCOMPOSICIÓN EN
FRACCIONES SIMPLES.

$$* = 2 \left[\int \left(1 + \frac{4}{t^2-4} \right) dt \right] = 2 \left[\int dt + 4 \int \frac{dt}{t^2-4} \right]$$

$$\int dt = t + c$$

$$* 4 \int \frac{dt}{t^2-4} =$$

$$\frac{1}{t^2-4} = \frac{1}{(t-2)(t+2)}$$

$$\frac{1}{(t-2)(t+2)} = \frac{A}{t-2} + \frac{B}{t+2}$$

$$\frac{1 \cancel{(t-2)} \cancel{(t+2)}}{\cancel{(t-2)} \cancel{(t+2)}} = \frac{A \cancel{(t-2)} (t+2)}{t-2} + \frac{B (t-2) \cancel{(t+2)}}{t+2}$$

$$1 = A(t+2) + B(t-2)$$

5 c REEMPLAZAMOS POR LA RAÍZ "2":

$$1 = A(2+2) + B(2-2)$$

$$1 = 4A$$

$$\boxed{A = \frac{1}{4}}$$

REEMPLAZAMOS POR LA RAÍZ "-2":

$$1 = \cancel{A(-2+2)} + B(-2-2)$$

$$1 = -4B$$

$$\boxed{B = -\frac{1}{4}}$$

$$= 4 \left[\int \frac{\frac{1}{4}}{t+2} dt + \int \frac{-\frac{1}{4}}{t-2} dt \right] = 4 \left[\frac{1}{4} \int \frac{dt}{t+2} - \frac{1}{4} \int \frac{dt}{t-2} \right] =$$

$$= \ln(t+2) - \ln(t-2) + C.$$

COMO EL REEMPLAZO ES $x+4 = t^2 \rightarrow t = (x+4)^{\frac{1}{2}} \rightarrow t = \sqrt{x+4}$

$$= \ln(\sqrt{x+4}+2) - \ln(\sqrt{x+4}-2) + C.$$

$$* = 2 \left\{ \sqrt{x+4} + \ln(\sqrt{x+4}+2) - \ln(\sqrt{x+4}-2) + C \right\}$$

$$243) \int \frac{\sqrt{x^3} - \sqrt[3]{x}}{6\sqrt[4]{x}} dx = \int \frac{x^{\frac{3}{2}} - x^{\frac{1}{3}}}{6 \cdot x^{\frac{1}{4}}} dx =$$

Exponentes: $\frac{3}{2} ; \frac{1}{3} ; \frac{1}{4}$ m.c.m. = 12

$$x = t^{12} \rightarrow t = \sqrt[12]{x}$$

$$dx = 12 t^{11} dt$$

$$= \int \frac{(t^{12})^{\frac{3}{2}} - (t^{12})^{\frac{1}{3}}}{6(t^{12})^{\frac{1}{4}}} \cdot 12 t^{11} dt =$$

$$= \int \frac{t^{18} - t^4}{6 t^3} \cdot 12 t^{11} dt = \frac{2}{1} \int \frac{t^{29} - t^{15}}{t^3} dt =$$

$$= 2 \int \left(\frac{t^{29}}{t^3} - \frac{t^{15}}{t^3} \right) dt = 2 \left[\int t^{26} dt - \int t^{12} dt \right] =$$

$$= 2 \left[\frac{1}{27} t^{27} - \frac{1}{13} t^{13} + C \right] = 2 \left[\frac{1}{27} (x^{\frac{1}{12}})^{27} - \frac{1}{13} (x^{\frac{1}{12}})^{13} + C \right] =$$

$$= \frac{2}{27} x^{\frac{27}{12}} - \frac{2}{13} x^{\frac{13}{12}} + C = \frac{2}{27} \sqrt[4]{x^9} - \frac{2}{13} \sqrt[12]{x^{13}} + C$$